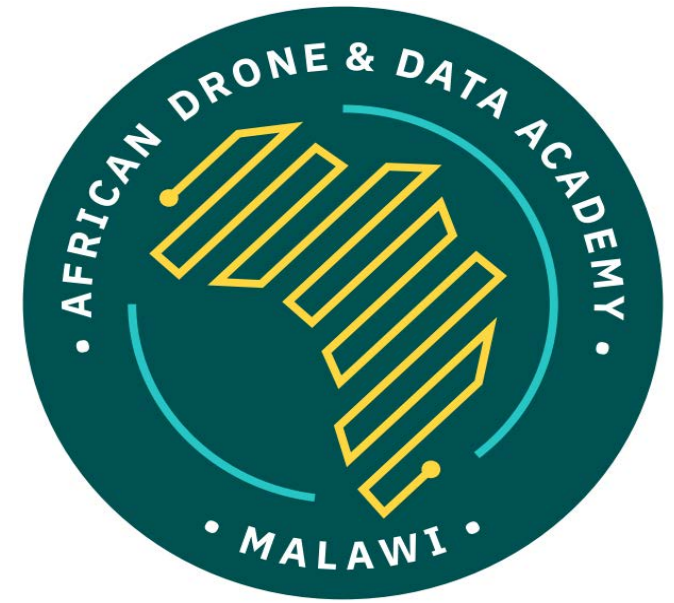




TOP Level 1

Day 2 Version A





TOP Level 1

Work Safety



Work safety: Occupational Health and Safety (OH&S)

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

- The need for safety cannot be underestimated in the operation of drones
 - Any **physical stress** can impact pilot performance
 - **Pilot safeguarding** reduces the likelihood of injury potential
 - **Areas of concern in OH&S include:**
 - Crew fatigue
 - Protection from the elements
 - Dehydration and heat related risks
 - Sun
 - Thunderstorms
 - Personal protective equipment (PPE)
 - Distractions and interference



OH&S: Crew Fatigue

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

- Fatigue is a significant risk factor to successful mission completion and overall safety.
 - Pilots that are fatigued do not perform as well and can miss obvious signals indicating a departure from safe flight
 - Hours of pilot rest are a major factor of fatigue

Recommended work/rest guidelines:

- *Maximum duty time - 12 hours per day (this includes any required meetings, standardization flights, ground labs, classes or any other assignment)*
- *Maximum duty time - 70 hours per week*
- *Minimum of 10 consecutive hours of rest during the 24-hour period that precedes the completion of the last activity.*

OH&S: Protection from the Elements

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

- Dehydration and heat-related risks:
 - Ensure a supply of drinking water is available during flight ops, and breaks are scheduled to access it
 - Effects of dehydration include:
 - Thirst
 - Headache
 - Fatigue
 - Cramps
 - Sleepiness
 - Dizziness
 - Heat exhaustion symptoms include the above, plus low blood pressure, nausea and low fever
 - Heat stroke symptoms include rapid heartbeat and rapid breathing, plus stroke-like symptoms of confusion, hallucinations, seizures and LOC



OH&S: Protection from the Elements

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

- Sun exposure can cause sunburn and it impacts the pilot's ability to see the aircraft
 - Recommend wearing a wide-brimmed hat and sunglasses to improve pilot comfort
- Thunderstorms and the potential hazard of lightning strikes should be monitored
 - Operations should be suspended during a lightning storm and at least 30 minutes after the last lightning strike



OH&S: Personal Protective Equipment

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

- Personal protective equipment (PPE) includes equipment that is worn to reduce the risk of RPA-specific hazards.
 - It is required for all operations of RPA within ADDA programs
- Minimum PPE includes:
 - Eye protection
 - Closed toe shoes
 - Sleeved shirts
- Optional equipment depending on the work environment:
 - High visibility vest
 - Gloves
 - Hearing protection



OH&S: Distractions and interference

- 7. SOPs
- 8. SMS
- 9. **Work safety**
- 10. Crew roles

- Care should be taken to minimize distractions and interference
- When operating in the proximity of people who are not part of the flight mission, measures should be taken to enhance the safety of the operating environment:
 - Barriers or markers should be put up to separate the flight crew from bystanders
 - Signage is recommended that explains the flight operations (to reduced questions asked of the crew)
 - It may be worthwhile to brief bystanders on the flight operations
 - Additional crew may be brought in to manage crowds
 - Select clothing or high visibility vests may be worn to distinguish the flight crew to others on site

The FAA has identified 12 common causes of human factors errors when working in aircraft maintenance

7. SOPs
8. SMS
9. **Work safety**
10. Crew roles

1. Lack of Communication:



- Failure to transmit, receive, or provide enough information to complete a task. Never assume anything.
- Only 30% of verbal communication is received and understood by either side in a conversation. Others usually remember the first and last part of what you say.
- Improve your communication-
 - Say the most important things in the beginning and repeat them at the end.
 - Use checklists.

2. Complacency



- Overconfidence from repeated experience performing a task
- Avoid the tendency to see what you expect to see-
 - Expect to find errors.
 - Don't sign it if you didn't do it.
 - Use checklists.
 - Learn from the mistakes of others.

3. Lack of Knowledge



- Shortage of the training, information, and/or ability to successfully perform.
- Don't guess, know-
 - Use current manuals.
 - Ask when you don't know.
 - Participate in training.

4. Distractions



- Anything that draws your attention away from the task at hand.
- Distractions are the #1 cause of forgetting things, including what has or has not been done in a maintenance task.
- Get back in the groove after a distraction-
 - Use checklists.
 - Go back 3 steps when restarting the work.

5. Lack of Teamwork



- Failure to work together to complete a shared goal.
- Build solid teamwork-
 - Discuss how a task should be done.
 - Make sure everyone understands and agrees.
 - Trust your teammates.

6. Fatigue



- Physical or mental exhaustion threatening work performance
- Eliminate fatigue-related performance issues-
 - Watch for symptoms of fatigue in yourself and others.
 - Have others check your work.

The FAA has identified 12 common causes of human factors errors when working in aircraft maintenance

- 7. SOPs
- 8. SMS
- 9. **Work safety**
- 10. Crew roles

7. Lack of Resources



- Not having enough people, equipment, documentation, time, parts, etc., to complete a task
- Improve supply and support-
 - Order parts before they are required.
 - Have a plan for pooling or loaning parts

8. Pressure



- Real or perceived forces demanding high-level job performance.
- Reduce the burden of physical or mental distress-
 - Communicate concerns.
 - Ask for extra help.
 - Put safety first.

9. Lack of Assertiveness



- Failure to speak up or document concerns about instructions, orders, or the actions of others.
- Express your feelings, opinions, beliefs, and needs in a positive, productive manner-
 - Express concerns but offer positive solutions.
 - Resolve one issue before addressing another

10. Stress



- A physical, chemical or emotional factor that causes physical or mental tension
- Manage stress before it affects your work-
 - Take a rational approach to problem solving
 - Take a short break when needed.
 - Discuss the problem with someone who can help.

11. Lack of Awareness



- Failure to recognize a situation, understand what it is, and predict the possible results.
- See the whole picture-
 - Make sure there is no conflict with an existing repair or modifications.
 - Fully understand the procedures needed to complete a task.

12. Norms



- Expect, yet unwritten, rules of behavior.
- Help maintain a positive environment with your good attitude and work habits-
 - Existing norms don't make procedures right.
 - Follow good safety procedures.
 - Identify and eliminate negative norms.



TOP Level 1

Crew Roles



The roles for flight personnel are defined in the flight operations manual

- FOM 3.5.4 discusses the roles and responsibilities for the crew during flight
 - The **remote pilot in command (RPIC)** is the final authority on all flight decisions and is the single responsible person regarding the flight mission
 - The RPIC ensures the RPA operation complies with all applicable regulations
 - The RPIC must have the ability to direct the RPA to ensure compliance with the applicable provisions of this chapter
 - The **remote pilot instructor (RPI)** offers training to the student pilots (SP)
 - This person trains the SP in a culture of safe flight operation
 - The RPI will take the student through a training syllabus which covers the elements of a Level 2 program aimed at the DCA RPA standards in Malawi

7. SOPs
8. SMS
9. Work safety
10. **Crew roles**



The roles for flight personnel are defined in the flight operations manual

7. SOPs
8. SMS
9. Work safety
- 10. Crew roles**

- **Visual observer (VO)** duties are as follows:
 - The VO is optional but recommended if personnel are available to assist
 - The RPIC must maintain visual line of sight to the aircraft, but the addition of a VO will enhance situational awareness and the safety of the flight mission
 - The VO must be able to effectively communicate:
 - The RPA location, attitude, altitude, and direction of flight;
 - The position of other aircraft or hazards in the airspace; and
 - The determination that the RPA does not endanger the life or property of another



Roles and responsibilities should be clearly documented so that there is no confusion about who is performing what task

7. SOPs
8. SMS
9. Work safety
- 10. Crew roles**

- **Student pilot (SP)**

- The Student pilot is the person receiving instruction from the RPI
- The SP works under direct supervision of the RPI and as such, will follow commands closely to ensure that flight operations are conducted safely and effectively
- The SP should use good judgement while commanding an aircraft that they are controlling
 - Any abnormal flight conditions should be brought to the attention of the RPIC immediately





TOP Level 1

Commercial Operations



Commercial flight operations in Malawi require a **Remotely Piloted Aircraft Operator Certificate (RPAOC)**

- No person shall engage in commercial, corporate and non-profit operations unless the operator holds a valid RPA Operator Certificate (ROC) issued by the Authority.
- The ROC shall authorise the operator to conduct RPA operations in accordance with the conditions and limitations detailed in the operations specifications attached to the ROC.
- The issuance of an ROC by the Authority is dependent upon the RPA operator demonstrating an adequate organisation, method of control and supervision of flight operations, training programme as well as ground handling and maintenance arrangements consistent with the nature and extent of the operations specified and commensurate with the size, structure and complexity of the organisation.
- A commercial RPA operator shall establish and implement a safety management system (SMS) in accordance with their operational requirements.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

Commercial flight operations: Application procedures for an RPA Operator Certificate

- A prospective operator applying to the Authority for an RPA Operator Certificate (ROC) shall submit an application in a form and manner prescribed by the Authority.
- A prospective operator shall make the application for an initial issue or reissue of an ROC at least 60 days before the date of the intended operation.
- No RPA shall be registered under more than one ROC.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Commercial flight operations: Issuance of a RPA Operator Certificate (ROC)

- The Authority may issue a ROC to the prospective operator if that applicant meet the following requirements:
 - Has its principal place of business and it is registered in Malawi.
 - Meets the requirements of the applicable regulations and for the holder of an ROC.
 - Is properly qualified and adequately staffed and equipped to conduct safe operations in commercial operations of the RPA.
 - Hold a security clearance issued by the relevant authority.
 - Has an operations manual as prescribed in IS16:63.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Operations manual, docs and records

- **An ROC holder shall develop and submit to the Authority for approval an Operations Manual as set out in the IS16:63**
- An ROC holder shall establish a system of record-keeping that allows adequate storage and reliable traceability of all activities developed, covering in particular
 - Lines of responsibility and accountability
 - Safety policy
 - Identification of aviation safety hazards encountered by the activities of the operator, assessment and mitigation of the associated risks, including taking actions and verifying their effectiveness.
 - Personnel training and competence
 - Quality, safety and security management records
 - The format of the records shall be specified in the ROC holder's operations manual
 - Records shall be stored for at least 2 years in a manner that ensures protection from damages, alteration and theft.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

IS 16:63: Operations Manual (FOM) Reqmts for the RPA Operator's Certificate

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

- Each ROC holder shall submit to the Authority for approval, an Operational Manual, Flight Manuals and maintenance manuals whose content commensurate with the size and scope of their intended operation. An Operations Manual shall include each item **set forth below** (on the next slide), unless otherwise approved by the Authority.

Part A - General (required in FOM)

1. INTRODUCTION

1. Purpose and scope of manuals
2. A statement that the manual complies with all applicable Authority regulations and requirements and with the terms and conditions of the applicable RPA operator certificate.
3. A statement that the manual contains operational instructions that are to be complied with by the relevant personnel in the performance of their duties.
4. List of manuals comprising operations manual
5. A list and brief description of the various operations manual parts, their contents, applicability and use.
6. Responsibility for manual content.
7. Responsibility for manual amendment.
8. List of effective pages.
9. Distribution of manuals and amendments.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

Part A - Safety management system (SMS) requirements for commercial ops

- The ROC holder shall establish a safety management system commensurate with the size of the organisation, structure and the complexity of its operations. IS16:63 Part A(2)
- The safety management system shall include:
 - A process to identify actual and potential safety hazards and assess the associated risks;
 - A process to develop and implement remedial action necessary to maintain an acceptable level of safety;
 - Provision for continuous and regular assessment of the appropriateness and effectiveness of safety management activities.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



Part A - General (required in FOM)

3. QUALITY SYSTEM

3.1 Description of quality system adopted

4. MANAGEMENT ORGANISATION

4.1 A description of the organisational structure including the general company organisation and operations department organisation. The relationship between the operations department and the other departments of the company. In particular, the subordination and reporting lines of all divisions, departments, etc., which pertain to the safety of the RPA operations, shall be shown.

4.2 Accountable Manager -duties and responsibilities.

4.3 Nominated personnel - functions duties and responsibilities.

4.4 RPA Pilot - duties and responsibilities.

4.5 Support personnel in the operation of RPA – duties and responsibilities

4.6 A description of the objectives, procedures and responsibilities necessary to exercise operational control with respect to flight safety.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

Part B - RPA operating information (required in FOM)

1. CREW INFORMATION

- 1.1 Flight team/crew composition
- 1.2 Qualification requirements of RPA pilot and support crew
- 1.3 Medical competencies
- 1.4 Operations of different types of RPA

2. OPERATIONS PROCEDURES OF RPA

- 2.1 Operating Limitations and conditions
- 2.2 Notification of intended operations to affected persons
- 2.3 Communications procedures
- 2.4 On site procedures and pre-flight checks

- 11. RPA operations
- 12. Flight performance
- 13. Meteorology
- 14. Navigation theory
- 15. Commercial ops**
- 16. Principles of flight
- 17. Human performance
- 18. Proficiency 1
- 19. Proficiency 2

Part B - RPA operating information (required in FOM)

3. RPA FLIGHT MANAGEMENT

- 3.1 Assembly and functional checks
- 3.2 Pre-flight checks
- 3.3 Normal flight procedures associated with relevant systems——
- 3.4 Inflight checks associated with relevant systems
- 3.5 Abnormal procedures associated with relevant systems
- 3.6 Emergency Procedures associated with relevant systems——

4. RPA USER MANUAL

- 11. RPA operations
- 12. Flight performance
- 13. Meteorology
- 14. Navigation theory
- 15. Commercial ops**
- 16. Principles of flight
- 17. Human performance
- 18. Proficiency 1
- 19. Proficiency 2

Part B - RPA operating information (required in FOM)

5. SECURITY: An operator shall ensure that policy and procedures in respect of the following aspects of security are addressed in its Operations Manual

- 5.1 RPA operator organisation and designation of a security coordinator;
- 5.2 Requirements for checks and searches of specific areas and accessible compartments of the interior and exterior of RPA;
- 5.3 Prevention of unauthorised access to remotely piloted aircraft and ground control stations;
- 5.4 Protection efforts pertaining to limiting the software and C2 links from forms of interference;
- 5.5 Response procedures for crew members and other staff or threats and incidents;
- 5.6 Special procedures
- 5.7 Crew member briefings concerning security/safety sensitive cargo loads
- 5.8 Additional security measures for special or more threatening situations
- 5.9 Reporting of security relating incidents to the Authority
- 5.10 Details on procedures and frequency on conducting background checks / criminal record checks
- 5.11 Details on security awareness and response procedure training.

- 11. RPA operations
- 12. Flight performance
- 13. Meteorology
- 14. Navigation theory
- 15. Commercial ops**
- 16. Principles of flight
- 17. Human performance
- 18. Proficiency 1
- 19. Proficiency 2

Part C - Areas, routes and aerodromes (required in FOM)

1. Areas of Operations
2. Operating site planning and assessment
3. Authorizations including site permissions

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2

Part D - Training (required in FOM)

1. Training syllabi and checking programs for RPA crew
2. Training syllabi and checking programs for RPA support crew
3. Training syllabi and programs for personnel other than crew
4. Recurrent training programs
5. Additional training requirements that individual clients specify for the proposed operations.

11. RPA operations
12. Flight performance
13. Meteorology
14. Navigation theory
- 15. Commercial ops**
16. Principles of flight
17. Human performance
18. Proficiency 1
19. Proficiency 2



TOP Level 1 Documentation



Aircraft documentation and records

- 20. Compliance
- 21. Documentation**
- 22. Aircraft logs
- 23. Maintenance logs
- 24. Battery logs

- All **flight operations manuals and documentation** are maintained on a Google Drive system and are accessible on the ADDA tablets and computers
- Manufacturer-specific information is accessible through this system
 - The Google Drive account will update each ADDA device when connectivity is available
 - The most recent documents are then downloadable in the field and are available at any time

ADDA S500 User Manual

1

ADDA User Manual for S500 V2 Quadcopter

Revision 1 – October 4, 2019

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TOP Level 1

Safety Policy



Safety policy

- The ADDA takes safety seriously, and for this reason has established a safety management system (SMS) to enforce best practices and create a safe operating culture
 - The SMS is accessible through a browser icon on ADDA devices - it guides safe operating practice at the ADDA
- The FOM also addresses risk management in Chapter 4 - it is also accessible through ADDA devices
- All reference manuals and publications are updated with internet connectivity, but the documents remain accessible without connectivity (in the field)

- 25. Safety policy
- 26. PPE equipment
- 27. PPE access
- 28. Job safety check
- 29. Log of hazards
- 30. Emerg resp plan





TOP Level 1

Emergency Response Plan



An emergency response plan (ERP) has been developed to improve the efficiency of an emergency response

25. Safety policy
26. PPE equipment
27. PPE access
28. Job safety check
29. Log of hazards
30. **Emerg resp plan**

- **Chapter 2 of the SMS** describes the ADDA emergency response plan
- The ERP addresses pilot responsibilities in the event of an emergency
- An **emergency response coordinator (ERC)** is designated as a focus to provide support in managing an emergency response



Emergency procedures quick-reference checklist (SMS §2.3):

1. If necessary, call emergency services.
2. Render first aid as necessary.
3. Contact the ADDA Head of Training or the ADDA Project Manager via the call out list.
4. Preserve accident/incident site to ensure aircraft wreckage is not tampered with.
5. Secure the Ground Control Station.
6. Once the flight crew and scene are secure/safe take pictures and begin documentation of the incident.

25. Safety policy
26. PPE equipment
27. PPE access
28. Job safety check
29. Log of hazards
- 30. Emerg resp plan**



Procedures in the event of an aircraft crash (SMS §2.6):

- **The crew and accompanying personnel following an aircraft crash are to:**

- protect themselves and others first
- make sure that the situation is dealt with in order to prevent further injury and damage
- get help from surrounding personnel and bystanders as well as notify any ADDA authority figures or emergency services, if required
- control the damage caused by the incident and prevent it from spreading
- record as much information about the aircraft crash as possible to identify the point of failure, learn from the incident, and avoid similar incidents in the future. This furthers the ADDA's goal of adapting our safety culture for changing circumstances and situations.

- 25. Safety policy
- 26. PPE equipment
- 27. PPE access
- 28. Job safety check
- 29. Log of hazards
- 30. Emerg resp plan

The FOM §3.4 addresses first aid and emergency response equipment

- The **emergency response coordinator** is designated by the Aviation Safety Officer, and will provide the following services at the flying site:
 - First aid kit that includes enough supplies to address most common injuries (cuts and minor lacerations, burns, dehydration)
 - A phone number of the nearest emergency response unit for transportation to a hospital
 - 2l of drinking water
 - Clean patient space to administer first aid
 - A bucket of sand for putting out fires

- 25. Safety policy
- 26. PPE equipment
- 27. PPE access
- 28. Job safety check
- 29. Log of hazards
- 30. Emerg resp plan**

Additional emergency response planning information is available in Chapter 2 of the SMS



TOP Level 1

Safety Training



Safety training for remote pilots occurs in multiple stages at the ADDA

31. Accident records
32. Accident reporting
- 33. Safety training**

- Students are assigned to read the SMS manual
- Students are assigned to read the FOM
- In the Level 1 curriculum, Sec II, Reference 7, **Project feasibility and safety assessments** are covered
 - The flight planning process takes the RPIC through a risk assessment analysis prior to flight
- Non-normal scenarios and emergency procedures are examined in Level 1, Sec II, Reference 7
 - Emergency response plan and an emergency quick reference checklist is presented



Safety training and procedures at the ADDA create a culture for safe flight operations

- 31. Accident records
- 32. Accident reporting
- 33. Safety training**

- Level 1, Sec II, Reference 8 covers the following safety areas:
 - The roles of internal and external safety committees
 - Accident and incident reporting, and the anonymous safety reporting system
 - Occupational health and safety, including:
 - Physical stress
 - Crew fatigue
 - Personal protective equipment (PPE)
 - Distractions and interference

